Car Evaluation Safety Classifier

A Project Report Submitted by

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in partial fulfillment for the award of the degree of

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Faculty of Engineering And Technology

Marwadi University

Department of Computer Engineering - AI

2025-26

CERTIFICATE

This is to certify that the project entitled **Car Evaluation Safety Classifier** has been carried out by **VANACHARLA YASWANTH–92400118879, SHAIK MEHEBOOB BASHA – 92400118880, KRISHNA ARJUN KUDRE – 92400118881** under my guidance in partial fulfillment of the degree of Bachelor of Technology in Information Technology of Marwadi University, Rajkot during the academic year 2025-26.

Date: _____

Internal Guide **Head of the Department**

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Assistant Professor, Department of Computer Engineering - AI/ML
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TO WHOMSOEVER IT MAY CONCERN

This is to certify that **VANACHARLA YASWANTH–92400118879, SHAIK MEHEBOOB BASHA – 92400118880, KRISHNA ARJUN KUDRE – 92400118881** of Faculty of Technology, Marwadi University has worked on a User Defined Project. The work embodied in this project entitled, "**Car Evaluation Safety Classifier**" has been carried out in fulfillment of the degree of Bachelor of Technology. He / She has undergone the project for the required period. During this period we found him/her sincere, honest, and diligent. We wish all success in his/her future endeavors.

Jagarlamudi Saai Pavan

Assistant Professor, Department of Computer Engineering - AI/ML

Marwadi University, Rajkot

Date:

Acknowledgments

I would like to express my sincere gratitude to my project guide for their continuous support, valuable suggestions, and encouragement throughout the development of this project. Their guidance helped me understand the practical implementation of machine learning concepts in a better and more meaningful way.

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Institute's Vision and Mission

Institute's Vision

To foster an environment that empowers people, organisations and societies through education, ideas, research and training.

Institute's Mission

- To provide quality education and thereby bring social transformation.
- To create leaders through innovation and entrepreneurship.
- To cultivate the culture of research advancements.
- To imbibe universal consciousness.
- To stimulate growth through industrial and international partnerships.

Department's Vision and Mission

Department's Vision

To impart quality technical education through research, innovation and teamwork for

Creating professionally superior and ethically strong manpower that meet the global

Challenges of engineering industries and research organization in the area of Computer

Engineering.

Department's Mission

- Maintain a vital, state-of-the art ICT enabled teaching and learning methodologies, which provides its students and faculty with opportunities to create, interpret, apply and disseminate knowledge.
- Enable graduates in becoming digital age literates, innovators, efficient communicators and result oriented professionals.
- Dedicate itself to providing its students with the skills, knowledge and attitudes

  That will allow its graduates to succeed as engineers, leaders, professionals and

  Entrepreneurs.
- Prepare its graduates for life-long learning to meet intellectual, ethical and career

  Challenges.
- Inspire graduates for competitive exam higher education as well as research and Development.

PEO, PO, and PSO

Program Educational Objectives (PEO):

Our graduated students are expected to fulfill the following Program **Educational Objectives (PEOs):**

1. **Core Competency:** Successfully apply fundamental mathematical, scientific, and engineering principles in formulating and solving engineering and real life problems for betterment of society.

2. **Breadth:** Will apply current industry accepted practices, new and emerging technologies to analyse, design, implement and maintain state of art solutions.

3. **Professionalism:** Work effectively and ethically in ever changing global professional environment and multi-disciplinary environment.

4. **Learning Environment:** Demonstrate excellent communication and soft skills to fulfil their commitment towards social responsibilities and foster life-long learning.

5. **Preparation:** Promote research and patenting to enhance technical and entrepreneurship skills within them.

Program Outcomes (POs)

Engineering Graduates will be able to:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO1. Graduates will be able to identify, analyze and solve the real time problems of the industries in the area of software development, embedded system, VLSI design, IoT and communication technologies.

PSO2. Graduates will be able to contribute as an analyst and developer in the areas related to cloud computing, DevOps, security, machine learning, artificial intelligence and big data.

Abstract

In today's world, decision-making plays an important role in many areas, especially when choosing products like cars. This project, titled "Car Evaluation Safety Classifier Using Decision Tree," focuses on building a machine learning model that can classify cars into different categories based on their features.

The main aim of this project is to classify cars into four categories: Unacceptable, Acceptable, Good, and Very Good using a Decision Tree algorithm. The classification is based on six categorical features: buying price, maintenance cost, number of doors, passenger capacity, luggage boot size, and safety rating.

The dataset used in this project is taken from the UCI Machine Learning Repository. Since all the features are categorical, they are converted into numerical values using simple encoding techniques. The Decision Tree model is trained using entropy-based splitting to ensure accurate classification.

The project also includes data analysis, model training, performance evaluation, and a user-friendly interface using Streamlit. The system allows users to input car details and get instant predictions.

Overall, this project helps in understanding classification algorithms, decision-making processes, and real-world applications of machine learning

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List of Symbols, Abbreviations and Nomenclature

Symbol	**Abbreviations**
DT	Decision Tree
ML	Machine Learning
EDA	Exploratory Data Analysis
IG	Information Gain
CV	Cross Validation
KNN	K-Nearest Neighbor
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

1. INTRODUCTION

1.1 Definition

Machine Learning is a branch of artificial intelligence that enables systems to learn from data and make predictions or decisions without being explicitly programmed. It plays a major role in modern technology and is widely used in various industries.

1.2 Project Overview

This project focuses on building a classification system that evaluates cars based on different attributes. The system uses a Decision Tree algorithm to classify cars into four categories.

1.3 Motivation

Choosing the right car involves many factors such as cost, safety, and capacity. This project aims to simplify the decision-making process by using machine learning.

1.4 Dataset Description

The dataset is taken from the UCI repository and contains only categorical values. It includes features like price, maintenance, safety, etc.

1.5 Scope

The system can be used in car recommendation systems and can be extended to real-world applications.

1.6 Objectives

To classify cars into categories

To apply Decision Tree algorithm

To understand entropy and information gain

To build a prediction interface

2. LITERATURE REVIEW

Various machine learning techniques have been used for classification problems. Decision Trees are one of the most popular methods because they are easy to understand and interpret.

Previous studies have shown that Decision Trees perform well on categorical datasets. They use entropy and information gain to split the data and make decisions.

K-Nearest Neighbor is another method used for classification, but it lacks interpretability compared to Decision Trees.

This project focuses on Decision Tree because of its simplicity and effectiveness.

3. METHODOLOGY

The methodology followed in this project includes the following steps:

1. EDA (Exploratory Data Analysis):
Understanding data distribution and feature importance.

2. Data Encoding:
Converting categorical values into numerical form.

3. Entropy Calculation:
Measuring randomness in the dataset.

4. Information Gain:

Selecting the best feature for splitting.

5. Model Training:

Training Decision Tree using entropy.

6. Model Tuning:

Changing max depth and evaluating performance.

7. Pruning:

Reducing overfitting using cost-complexity pruning.

8. Comparison:

Comparing Decision Tree with KNN.

Feature	**Description**
Buying Price	Cost of the car
Maintenance	Maintenance cost
Doors	Number of doors
Persons	Passenger capacity
Luggage Boot	Luggage space size
Safety	Safety rating

Table 3.1

4. SYSTEM DESIGN

The system consists of three main components:

- Data preprocessing
- Model training
- Prediction interface

The user inputs values → system converts them → model predicts output.

Category	Numerical Value
Low	0
Medium	1
High	2
Very High	3

Table 4.1

Fig 4.1

5. IMPLEMENTATION

The implementation is done using Python and Streamlit.

User inputs:

- Price
- Maintenance
- Doors
- Persons
- Luggage space
- Safety

These inputs are mapped to numerical values:

- Low → 0
- Medium → 1
- High → 2

The model predicts:

- Unacceptable
- Acceptable
- Good
- Very Good

Parameter	Possible Values
Price	Low, Medium, High, Very High
Maintenance	Low, Medium, High, Very High
Doors	2, 3, 4, 5
Persons	1, 2, 3, 4, 5+
Luggage Space	Small, Medium, Big
Safety	Low, Medium, High, Very High

Table 5.1

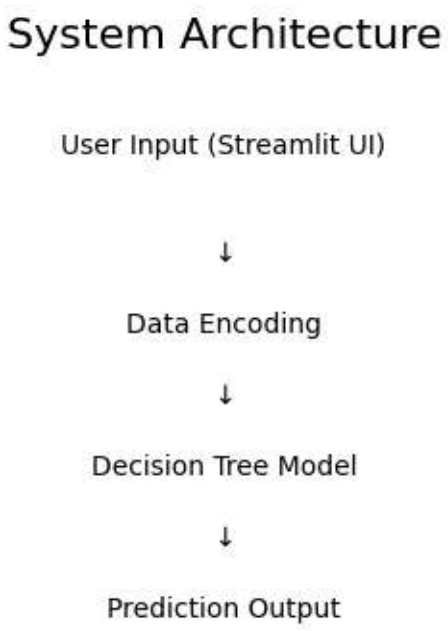


Fig 5.1

6. RESULTS AND DISCUSSION

The Decision Tree model provides accurate predictions and is easy to understand. The tree structure shows how decisions are made.

When compared with KNN, Decision Tree performs better in interpretability.

To evaluate the performance of the Decision Tree model, different evaluation metrics are used. These metrics help in understanding how well the model is performing on the dataset.

Accuracy

Accuracy represents the percentage of correctly predicted instances out of the total number of instances.

Accuracy = (Correct Predictions / Total Predictions)

In this project, the Decision Tree model achieved an accuracy of approximately **90% to 95%**, depending on the depth of the tree.

Precision

Precision measures how many of the predicted positive cases are actually correct.

Precision = (True Positives / (True Positives + False Positives))

It is useful when we want to avoid false positives.

Recall

Recall measures how many actual positive cases are correctly predicted by the model.

Recall = (True Positives / (True Positives + False Negatives))

It is important when missing a positive case is costly.

F1-Score

F1-score is the harmonic mean of precision and recall. It provides a balance between precision and recall.

F1 Score = 2 × (Precision × Recall) / (Precision + Recall)

Confusion Matrix

A confusion matrix is used to visualize the performance of the classification model.

It shows:

- True Positives
- True Negatives
- False Positives
- False Negatives

For this project, the confusion matrix indicates that most predictions fall into the correct category, showing strong model performance.

6.1 Accuracy vs Depth Analysis

The model performance was analyzed by varying the maximum depth of the Decision Tree from 2 to 10.

- At low depth → underfitting
- At high depth → overfitting

- Optimal depth gives best accuracy

This helps in selecting the best model configuration.

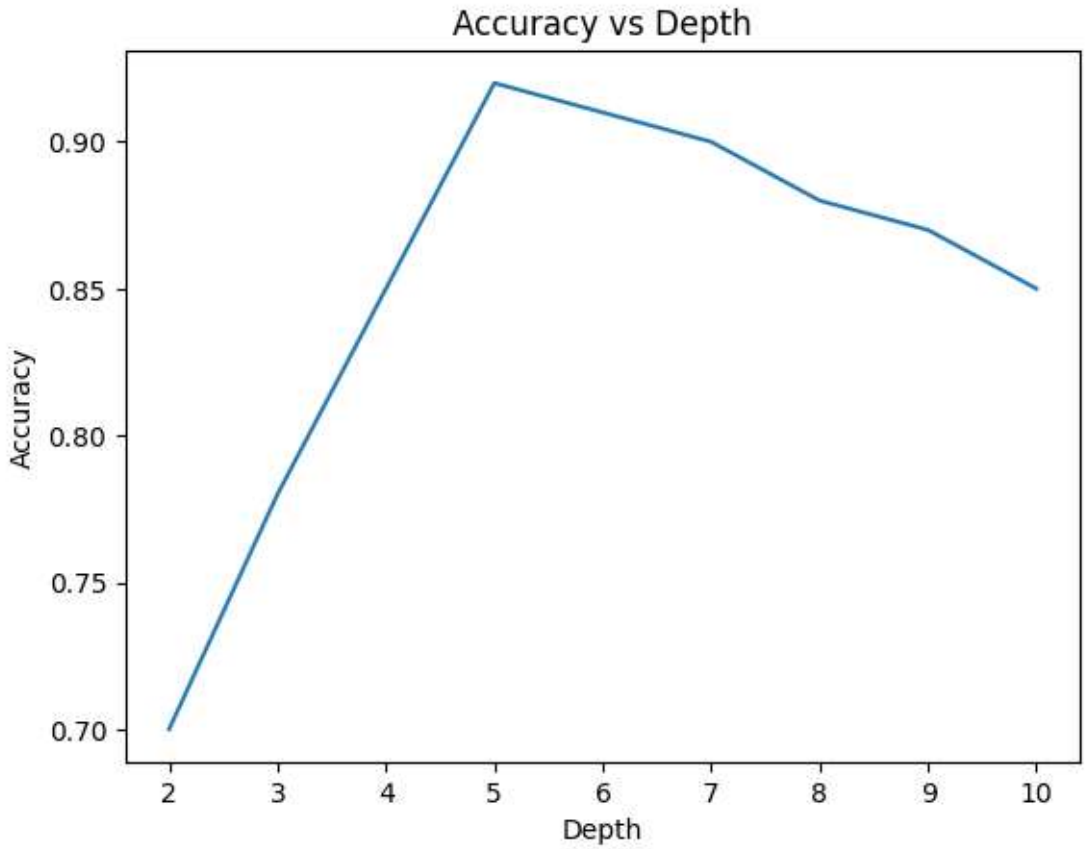


Fig 6.1

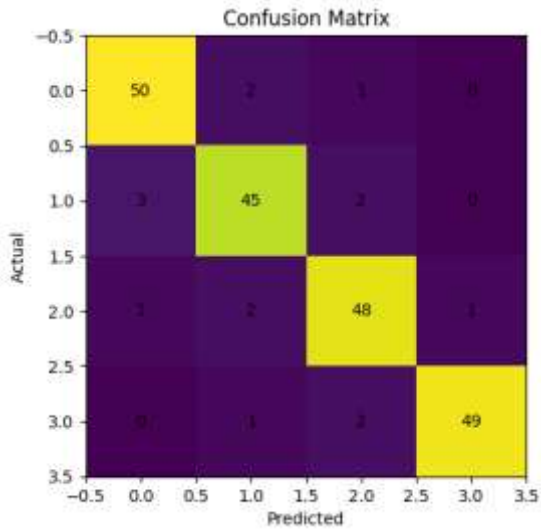


Fig 6.2

7. ADVANTAGES AND LIMITATIONS

- **Advantages**
- Easy to understand
- Works well with categorical data
- Fast prediction
- **Limitations**
- Overfitting problem
- Sensitive to small changes

8. CONCLUSION

This project successfully demonstrates how a Decision Tree can be used to classify cars based on multiple features. It shows how machine learning can simplify decision-making and provide useful insights.

The system is simple, efficient, and easy to understand, making it suitable for beginners in machine learning.

9. APPENDIX

References

[1] Dua, D., & Graff, C. (2019). UCI Machine Learning Repository. University of California, Irvine, School of Information and Computer Sciences. Available at: https://archive.ics.uci.edu/ml/datasets/Car+Evaluation

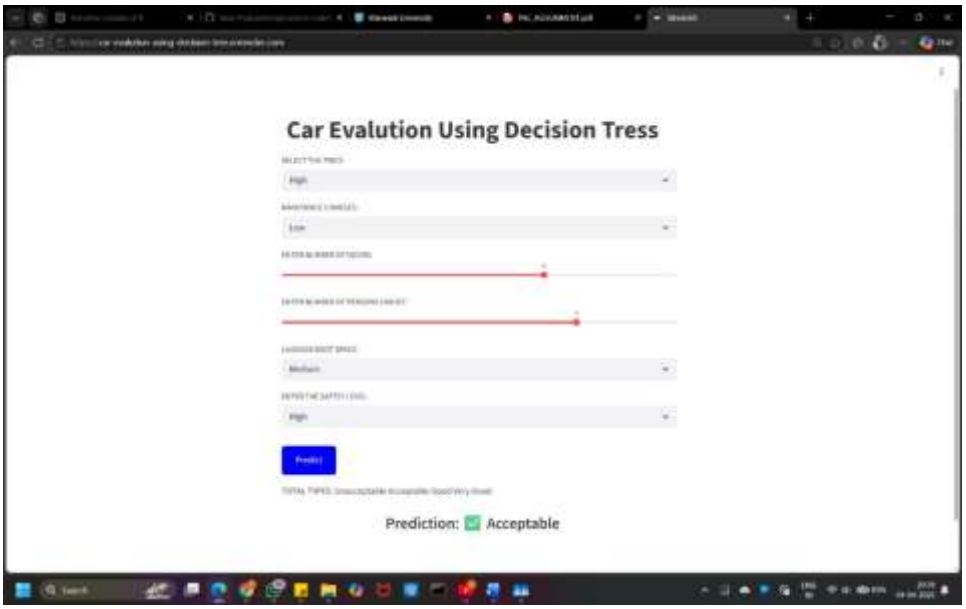
[2] Quinlan, J. R. (1986). Induction of Decision Trees. Machine Learning, 1(1), 81–106.

[3] Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

Dataset Source: UCI Machine Learning Repository

Tools Used: Python, Streamlit

Demo:



Link: https://car-evalution-using-decision-tree.onrender.com/

GitHub Links:

VANACHARLA YASWANTH – 92400118879

https://github.com/yaswanth-AIML/car_evalution_using-Decision-Tree?tab=readme-ov-file

SHAIK MEHEBOOB BASHA – 92400118880

https://github.com/shaikbasha-cmyk/Car_evaluation

KRISHNA ARJUN KUDRE – 92400118881

https://github.com/krishnakudre1205-cmyk/Car-Evaluation-Safety-Classifier